Explainable Accounting-Fraud Detection

Extending Bao et al. (Journal of Accounting Research, 2020) with scale-free features, firm history and SHAP

Final Year Project · 2026

ROADMAP

What this talk covers

1. **Project workflow**
   From raw data to an explained score
2. **The base paper**
   What Bao et al. (2020) did
3. **Rebuilding the paper**
   Same data, same settings, same result
4. **Literature review**
   Where the field stood
5. **Our improvements**
   Five gaps, five fixes
6. **Results**
   What improved, and by how much
7. **Future scope**
   What to do next, and how

From raw filings to an explained fraud score



Data

146k firm-years, 1991–2014, SEC fraud labels

Features

133 scale-free ratios and ranks + 7 market signals

Models

LightGBM, XGBoost, logistic regression, Transformer

Evaluation

Train on the past, test on a later year

Explain

SHAP: why each firm was flagged

Deploy

Calibrated probability and a scoring script



The feature pool is fixed, not re-selected each year. This is a tested design choice, not a shortcut: the next two slides give the argument and the evidence.

Why the features are fixed, not dynamic

1

Chosen once, on validation years

Picked as a set on 1999–2001 folds. The test years never influenced the choice.

2

Dynamic options were tested fairly

12 selection and generation methods ran inside every training window, under rules fixed before any run. None cleared the bar.

3

Fixed means stable and explainable

Every year's model uses the same features, so SHAP explanations can be compared and audited across years.

Direct test on the test years

+0.0008

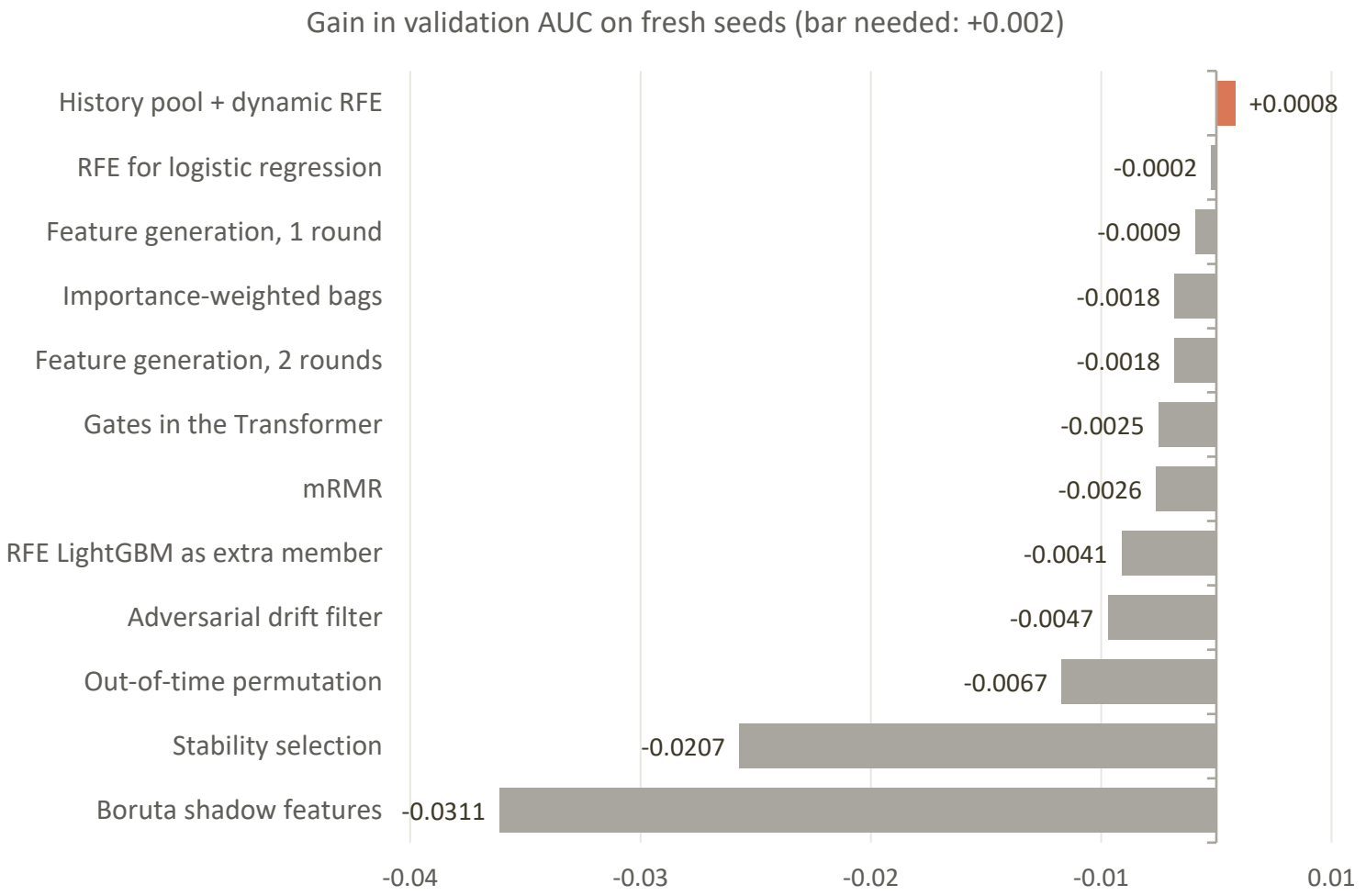
AUC change from dynamic selection. 95% CI −0.004 to +0.006: no measurable difference.

59–200

features kept for the same year, depending only on the random seed.

Dynamic selection adds cost and instability without a measurable gain.

Every dynamic method fell short of the bar



The bar
+0.002 AUC on fresh seeds, fixed before any run

Best result
+0.0008, below the bar

Why it fails
~960 fraud cases are too few to pick among hundreds of features

Decision: keep the fixed pool

Bao et al. (2020): machine learning finds fraud

Bao, Ke, Li, Yu & Zhang
Journal of Accounting Research, 58(1), 2020

Goal
Rank U.S. public firms by their risk of SEC fraud enforcement (AAERs), one year ahead

Method
RUSBoost: boosted trees with random under-sampling, built for rare events

Finding
Raw items beat expert ratios, and the model beats earlier benchmarks

28
raw financial items as inputs

0.66%
of firm-years are fraud

≈0.72
AUC on 2003–2008 test years

2 yrs
gap between training and test

How the paper tests: train on the past, predict the future

Expanding window, one row per test year



Expanding window

Retrain for every test year on all earlier data

2-year gap

Recent fraud is not public yet, so year T−1 is left out

Serial-fraud masking

A case running into the test year is removed from training

We rebuilt the paper in Python and matched it

Data	146,045 firm-years from 18,444 firms, 1991–2014
Fraud labels	964 fraud firm-years in 412 SEC cases
Model	RUSBoost: 300 trees, leaf size 5, learning rate 0.1
Sampling	Non-fraud under-sampled to 1:1 in every round
Tuning	Train 1991–1999, validate on 2001

Paper reports

≈0.72 AUC

Our rebuild

0.705–0.724

Reproduced: a fair baseline for every later gain

Baseline used from here on: 0.705, the paper's own tree settings.

Where the field stood before this project

2010 ≈0.63 AUC

Cecchini et al.
Support vector machine with a financial kernel

2011 ≈0.69 AUC

Dechow et al.
F-score: logistic model on 14 expert ratios

2017 used here

Lundberg & Lee
SHAP: consistent, per-prediction explanations

2020 ≈0.72 AUC

Bao et al.
RUSBoost on 28 raw items beats both benchmarks

2021 future work

Bertomeu et al.
Audit and governance data add signal

2023 tested here

Zhang et al. (OpenFE)
Automatic feature generation

Five gaps in the paper, five fixes

Scale bias
Raw dollar items let firm size drive risk

Scale-free ratios ranked within each year

Black box
No reason given for a flag

SHAP explanation from every model

No decision step
A ranking, with no cutoff

Budgets, lift and calibrated probabilities

Thin statistics
No confidence intervals

Bootstrap clustered by fraud case

Not deployable
No way to score new firms

Deployable model and scoring script

Features that measure risk, not firm size

Removed

The 28 raw dollar items, which mostly measure size

Added

Beneish M-score parts, Altman Z, accrual, growth and financing-need ratios

Ranked

Each ratio ranked within its own year, so 1991 and 2014 are comparable

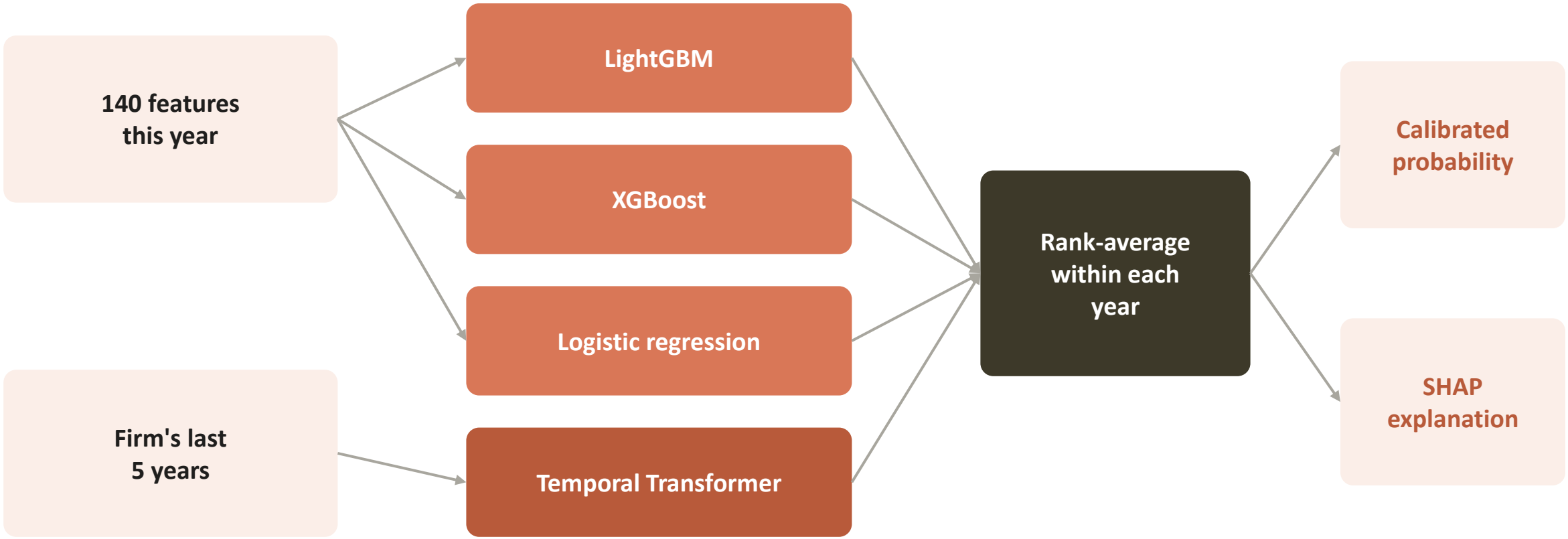
Firm history

A Transformer reads each firm's last 5 years

Market signals

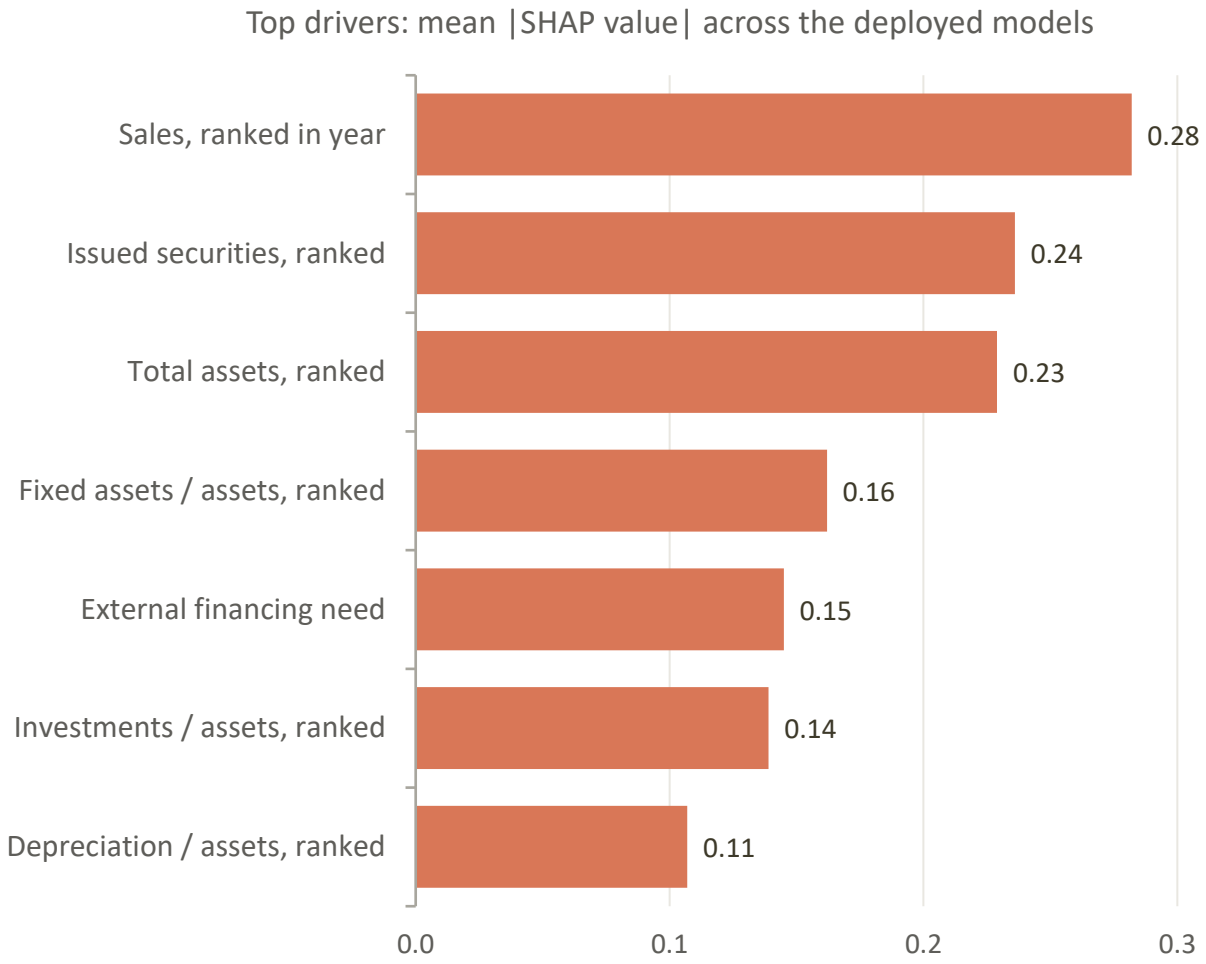
7 ranks of market-value and share movement

Four models, one ranked and explained score



Tuned only on data before 2003; the paper's model is never a member.

Every flag comes with its reasons



All four models explained
In the same log-odds units, not a stand-in model

Recent years weigh most
The Transformer puts 38% of attribution on the latest year

Audited
7 of 8 sanity checks passed; its global ranking is indicative

Headline results, test years 2003–2012

0.705 → 0.748

ROC AUC
Paper baseline to our model

9.8% → 15.1%

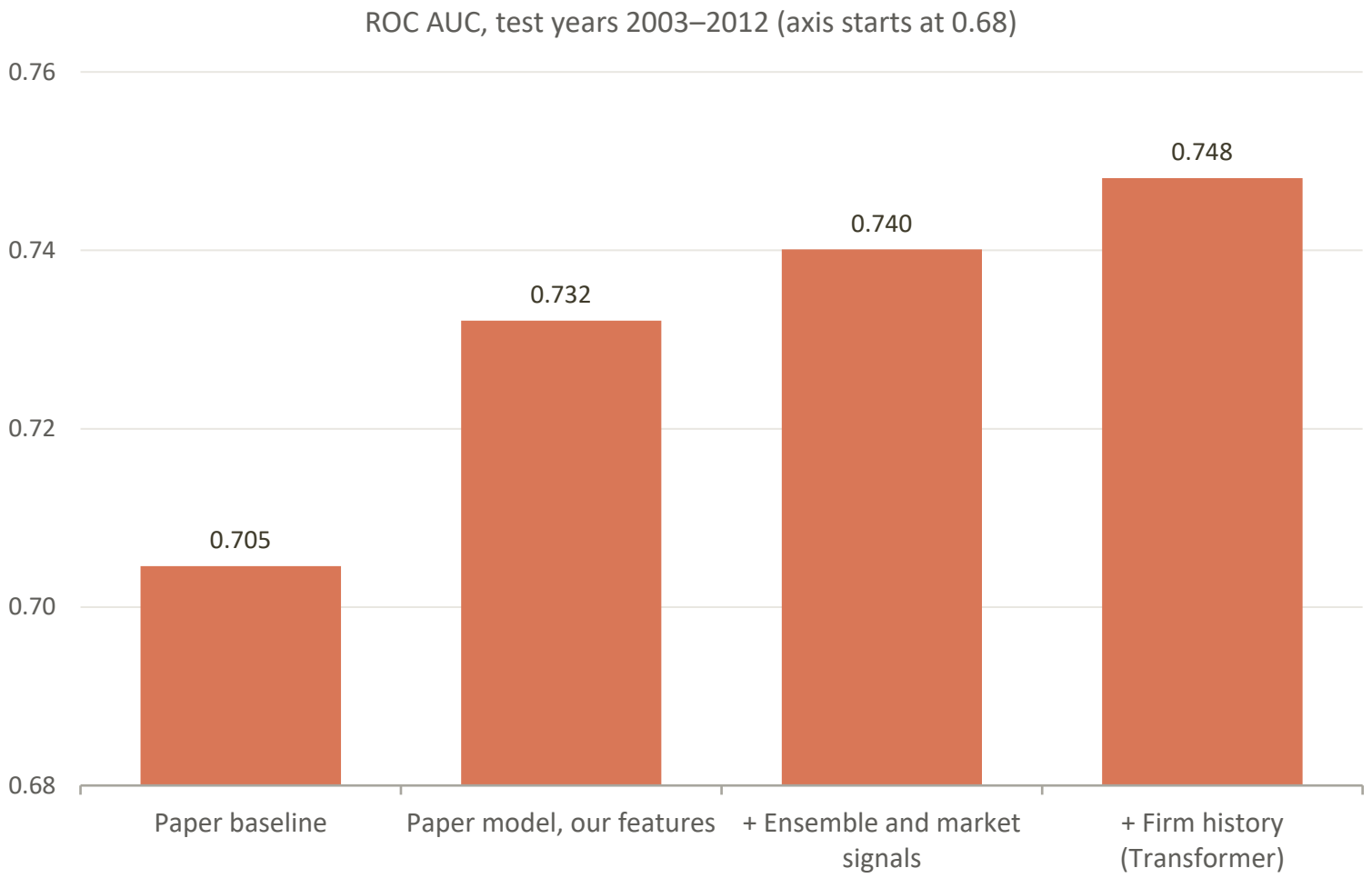
Frauds caught in the top 5%
54% more than the baseline

1.9 per 100

Frauds found per 100 reviews
At a 1% budget: 2.7× random

AUC and top-5% gains are significant under a bootstrap clustered by fraud case (1,000 resamples).

Where the gain comes from



+0.027

from the features alone

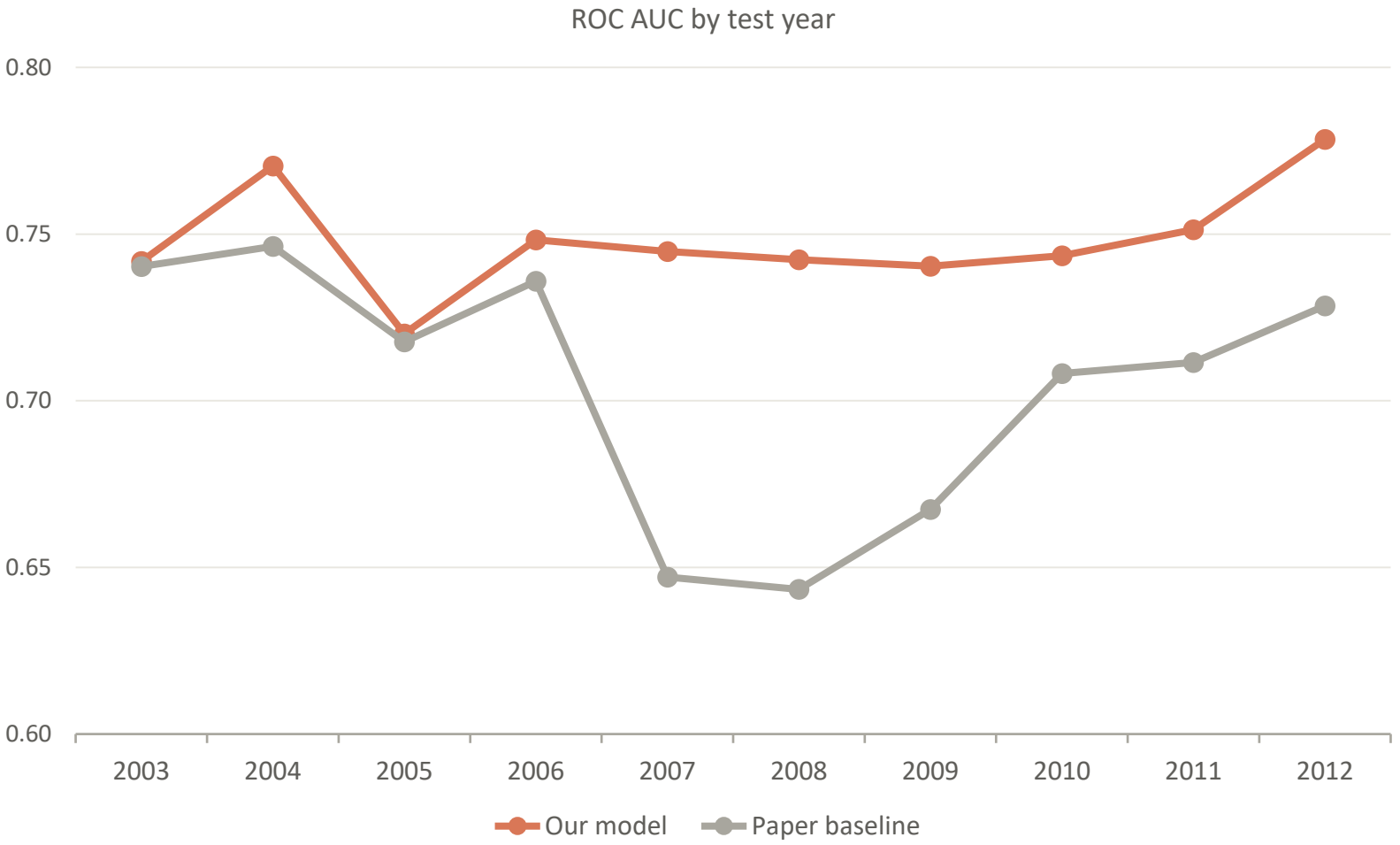
+0.008

from the ensemble and market signals

+0.008

from firm history

Ahead in every scored year



10 of 10 years
Our model is ahead in every scored year

Crisis years
Baseline drops to 0.64–0.67 in 2007–2009; ours stays near 0.74

2013–2014 left out
Fewer than 15 frauds each, too few to score

What the results do not show

A ranking tool, not a detector

About 98 of every 100 flagged firms are not labelled fraud. It raises an investigator's hit rate 2.7×.

Firm size still matters

Sales and assets rank highest in SHAP, even without raw dollar items.

Why not 0.8?

Weaker test rules add about 0.09 AUC on the same models. We keep the strict rules.

Small last step

The last +0.005 is promising, not confirmed after correcting for multiple tests.

What to do next, and how

WHAT	HOW
Audit and governance data	Add audit fees, auditor changes and board data from WRDS; retest under the same rules
Size-neutral model	Drop size features or model within size buckets; check that precision at 1% holds
Cost-based cutoff	Set the review budget from the cost of a review against the loss avoided per fraud caught
Label-delay modelling	Weight recent years by the chance their frauds are already found (survival model)
A fresh test period	Score 2015+ filings once their labels mature, for a test no model has seen

Thank you

Questions are welcome

- Reproduced the JAR 2020 baseline
- Raised ROC AUC from 0.705 to 0.748, with explanations
- Kept the features fixed after testing 12 dynamic alternatives